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GreenScore AI: A Circular Economy-Based Sustainability Scoring and Resource Optimization Framework for Smart Farming

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01 INTRODUCTION

Agriculture is responsible for significant consumption of water, energy, fertilizers, and other natural resources.

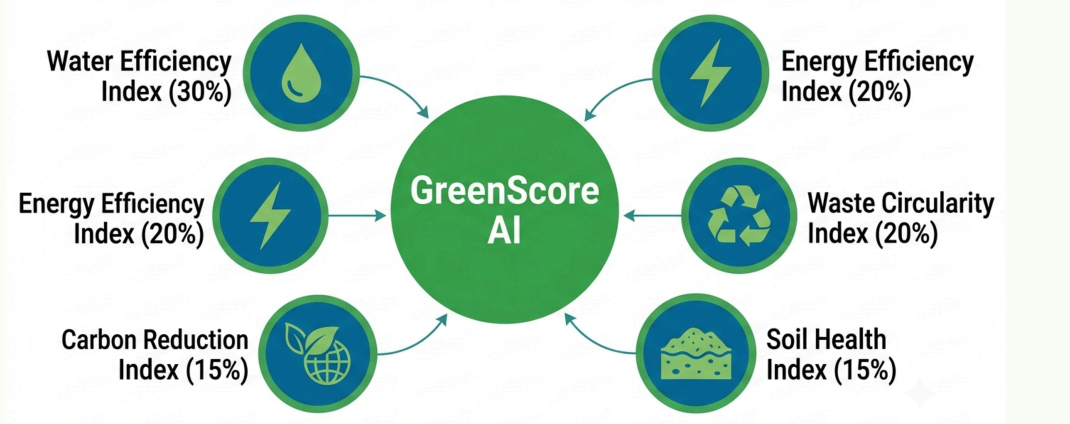
- Conventional farming systems often follow a linear resource-use model in which valuable resources are consumed and discarded with limited reuse. Such practices contribute to resource depletion, greenhouse gas emissions, and increased production costs.
- The circular economy offers an alternative approach by promoting resource recovery, recycling, and efficient utilization.

02 OBJECTIVES

- Develop an AI-based circular intelligence framework for evaluating sustainability in smart farming.
- Design a comprehensive Green Sustainability Index (GSI) by integrating multiple circular economy indicators.
- Develop a predictive decision-support system that recommends resource optimization strategies using machine learning.
- Promote zero-waste farming through intelligent monitoring of water, energy, carbon emissions, soil health, and agricultural residues.

03 METHODOLOGY

GreenScore AI is an AI-enabled decision support framework that integrates IoT sensors, cloud computing, machine learning, and circular economy principles to evaluate and optimize farm sustainability. The framework continuously collects field data, calculates a Green Sustainability Index (GSI), predicts future sustainability performance, and recommends resource-efficient farming practices.



$$GSI = 0.30 \times WEI + 0.20 \times EEI + 0.20 \times WCI + 0.15 \times CRI + 0.15 \times SHI$$

Figure 1. Green Sustainability Index (GSI) Framework

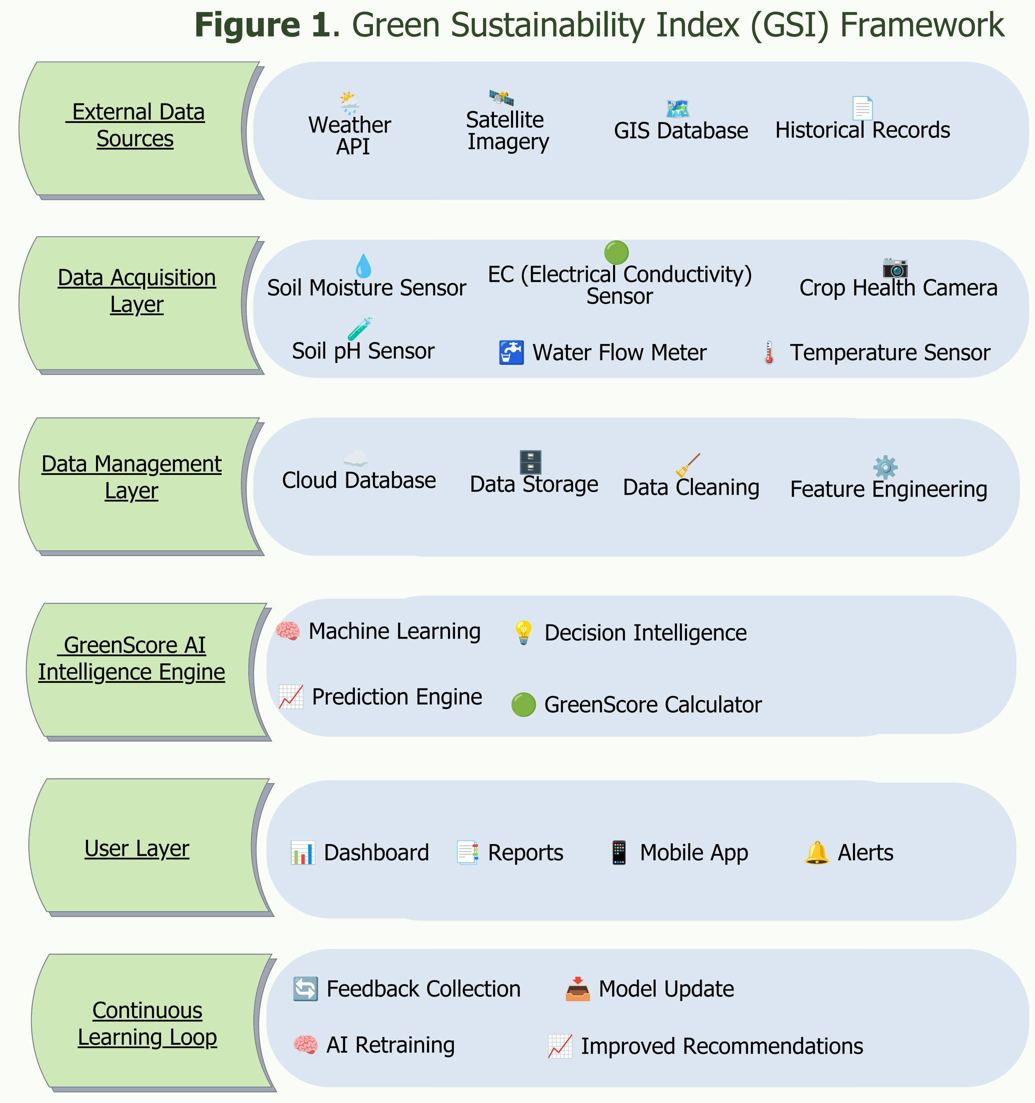


Figure 2. GreenScore AI System Architecture

04 RESULTS

The proposed GreenScore AI framework is expected to improve sustainability assessment by integrating multiple resource indicators into a unified Green Sustainability Index (GSI). Unlike conventional monitoring systems that evaluate resources independently, GreenScore AI provides continuous monitoring, predictive analytics, and AI-driven recommendations for circular resource utilization

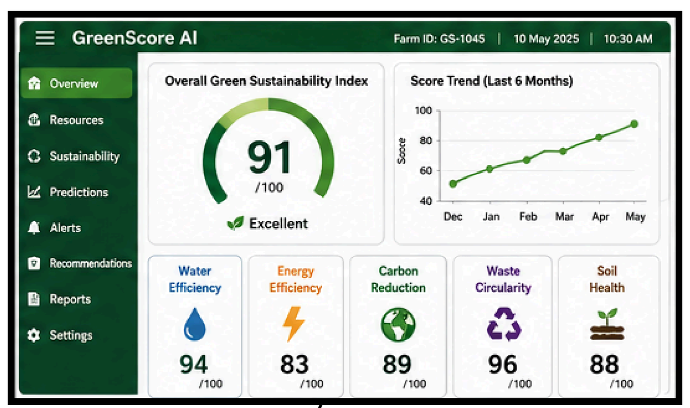


Figure 3. GreenScore AI Dashboard Conceptualised

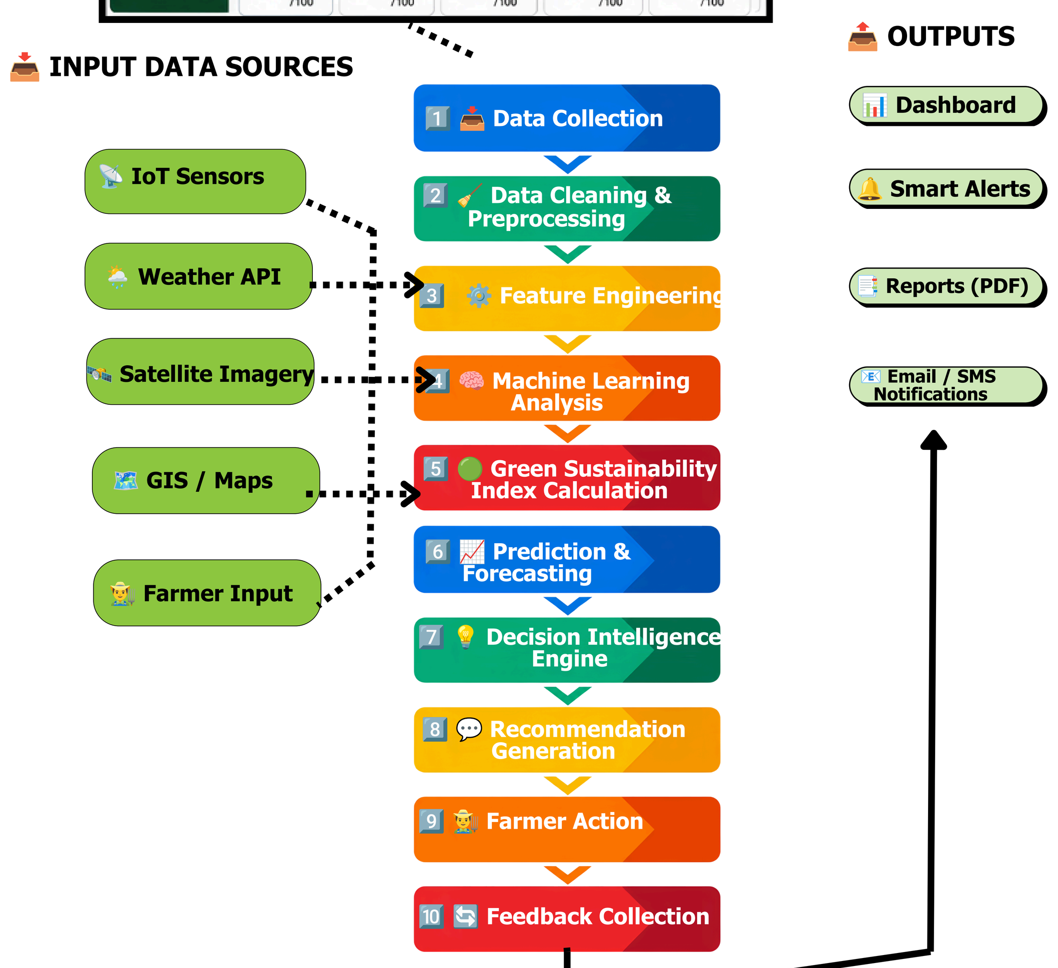
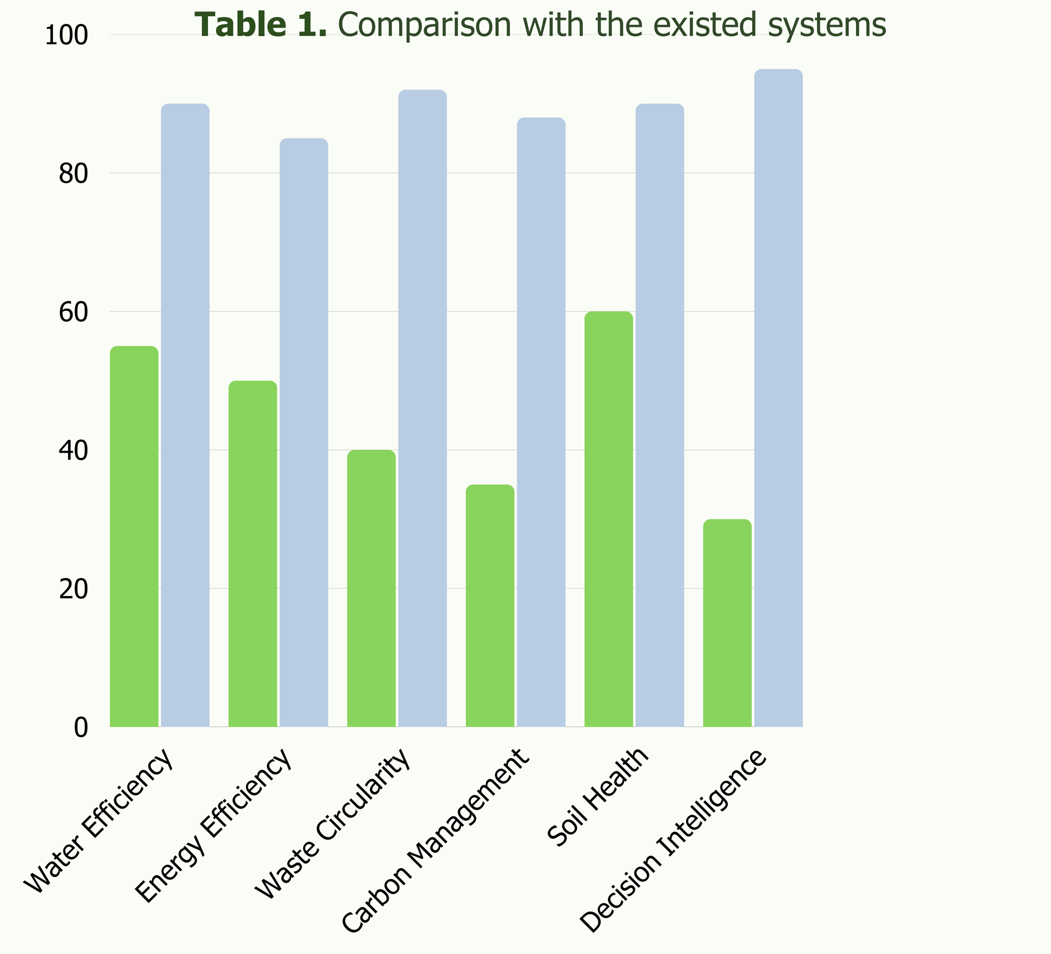


Figure 4. AI Decision Intelligence Workflow

Feature / Capability	Conventional Farm Monitoring	Farm ERP Systems	GreenScore AI (Conceptualised)
IoT Sensor Integration	0	Limited	✓✓
Sustainability Scoring	✗	✗	✓✓
Circular Economy Assessment	✗	✗	✓✓
Carbon Monitoring	Limited	✗	✓✓
Predictive Analytics	Limited	Limited	✓✓
Resource Optimization	Partial	Partial	✓✓
Continuous Learning	✗	✗	✓✓
Decision Intelligence	✗	✗	✓✓
User-Friendly Dashboard	Partial	Partial	✓✓
Multi-Source Data Integration	Partial	Partial	✓✓

Table 1. Comparison with the existed systems



Graph1. Conceptual comparison of sustainability dimensions between conventional farm management and the proposed GreenScore AI framework.

06 DISCUSSIONS

- GreenScore AI integrates AI, IoT, and Circular Economy principles into a single sustainability assessment framework.
- The Green Sustainability Index (GSI) provides a comprehensive evaluation of farm resource efficiency.
- AI-driven recommendations support predictive decision-making for water, energy, waste, and carbon management.
- The framework promotes resource optimization, reduced environmental impact, and climate-smart farming.
- The modular design enables future integration with real-time IoT systems and diverse agricultural environments.

Figure.7. Smart irrigation systems improve water-use efficiency through precise and controlled water application

- ✓ AI Recommendation
- ✓ Reduce Irrigation
- ✓ Save Water
- ✓ Lower Energy Use
- ✓ Lower Carbon Emissions
- ✓ Higher GreenScore

Figure 8 & 9. Recycling crop residues into compost promotes nutrient cycling and supports circular agricultural practices

06 CONCLUSIONS

- GreenScore AI introduces an integrated AI-driven framework that combines IoT, machine learning, and circular economy principles to provide a unified sustainability assessment for smart farming.
- The proposed Green Sustainability Index (GSI) enables intelligent resource optimization and predictive decision support, helping improve water, energy, waste, carbon, and soil management within a single platform.
- The framework has strong potential to advance circular agriculture, enhance resource efficiency, reduce environmental impacts, and support climate-resilient and sustainable farming systems

GreenScore AI redefines sustainable agriculture by converting farm data into intelligent decisions that maximize resource efficiency and advance circular farming.

— KEY FINDING

07 FUTURE WORK

- Validate GreenScore AI through multi-location field trials using real-time IoT data.
- Integrate satellite imagery, drones, and digital twin technologies to improve prediction accuracy.
- Develop a scalable cloud and mobile platform through collaborations with academia, industry, and farming communities

08 REFERENCES

1. Wolfert, S., Ge, L., Verdouw, C. & Bogaardt, M. J. Big Data in Smart Farming – A Review. Agricultural Systems 153, 69–80 (2017).
2. Liakos, K. G., Busato, P., Moshou, D., Pearson, S. & Bochtis, D. Machine Learning in Agriculture: A Review. Sensors 18(8), 2674 (2018).
3. Zhang, Y., Wang, G., Wang, X. & Liu, H. Artificial Intelligence Applications in Smart Agriculture: A Review. Computers and Electronics in Agriculture 210, 107936 (2023).
4. Ellen MacArthur Foundation. Completing the Picture: How the Circular Economy Tackles Climate Change. Ellen MacArthur Foundation (2019)

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